Some of the Design Parameters for Industrial-Grade Appliance Displays

Typical screen size	Device	Number of touch sensor nodes (max.)	Packages	Interfaces	HID (Windows)	Passive stylus
2.5cm-10.2cm (1"-4")	MXT144U	144 (12X×12Y)	QFN38 & WLCSP36	I2C	No	No
7.6cm-17.8cm (3"-7")	MXT336U	336 (14X×24Y)	QFN56			Yes
12.7cm-25.4cm (5"-10")	MXT640U	640 (32X×20Y)	UFBGA88		Yes	
20.3cm-30.5cm (8"-12")	MXT1066T2	1066 (41X×26Y)	UFBGA144 & UFBGA117			
25.4cm-40.6cm (10"-16")	MXT1664T3	1664 (32X×52Y)	UFBGA162 & UFBGA136	I2C & USB		
33.0cm-61.0cm (13"-24")	MXT2952T2	2911 (41X×71Y)	UFBGA162			

like moisture, bare finger, stylus and glove rather than automatically detecting and adjusting settings on the fly for natural, intuitive user interactions in all environments.

In such applications, the user interface tends to be simpler with larger buttons and needs to be incorporated at the design stage of the touch screen, for appliances that take advantage of glove support.

Choosing the right touch controller/sensor/screen

Depending on the size of the kitchen or laundry appliance, there are a variety of screen sizes that can be used ranging from 7.6cm (3-inch) displays for coffee makers to 12.7cm (5-inch) for microwave, stovetops and washing machines and up to 55.9cm (22-inch) and larger for refrigerators and freezers.

In the appliance supply chain, the chip supplier provides the chip and the design of the sensor in partnership with a sensor supplier. Together they perform the system integration and typically a module/display manufacturer integrates the system including the touch sensor and the touch interface, which is then provided to the appliance manufacturer (see Fig. 3). This is an example of how a semiconductor supplier today not only provides just the chip but also services such as tuning the chip, making it easier to use in the entire supply chain.

Operating in the industrial -40°C to 85°C temperature range, the standard IC ships with standard firmware and is able to address a variety of display sizes and different appliance manufacturer requirements. Touch screen designs are more scalable, thanks to the wide range of screen size options available in the appliance touch screen controller family. The ready availability of the right screen size reduces design time and lowers system and development costs. Table here shows some of the design parameters for industrial-grade appliance displays.

In addition to screen size, other parameters can impact the choice of the chosen controller, especially if there is an overlap in screen sizes.

One final consideration is electromagnetic compliance (EMC). Obviously, the design must support EMC. Then testing must verify that it achieves the desired results for both conducted and radiated emissions.

Fig. 4: Microchip's ATEVK-MXT2952T2 evaluation kit contains a dedicated sensor with a flex connector and an electronic control board

The design touch

For early engagement and improved awareness of touch screen capabilities, a designated evaluation kit is available for each of the controllers in the appliance touch screen family. The kit includes a printed circuit board (PCB) with a touch screen controller that has a passive flexible printed circuit (FPC) tail that connects the touch IC to a touch sensor on a glass/plastic lens. The kit connects to the host PC via USB and includes all necessary cables, software and documentation.

Used with the maXTouch Studio Development System Integrated Development Platform (IDP), a full software development environment (a free download from the Web), the evaluation kit enables appliance designers to develop and debug appliance touch controllers. Fig. 4 shows what appliance manufacturers can expect to find in an evaluation kit.

The final touch

IoT is here and now, appliance manufacturers need to think about improvements that can benefit the end-user. Touch screen enabled appliances must be equipped to read or input information in any context. Appliance manufacturers need to work with an IC supplier, or touch screen or module supplier who in turn works with an IC supplier, where touch controllers are designed specifically for appliance applications. With the right touch controller, appliances can provide internet connectivity and have noise immunity, moisture immunity and glove support. EFY